Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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1. **Show that, for any square matrix A, the matrix $A + A^T$ is symmetric.**

   By definition, a matrix A is symmetric if and only if $A = A^T$.

   Let $A \in \mathcal{M}_n(\mathbb{K})$.
   Note that $A^T \in \mathcal{M}_n(\mathbb{K})$, so $A + A^T$ is defined.

   Using the properties of transposition:

$$\begin{aligned}(A + A^T)^T &= A^T + (A^T)^T \\ &= A^T + A \\ &= A + A^T \implies \\ \implies &\boxed{(A + A^T)^T = A + A^T}.\end{aligned}$$

2. Final item. Remove when done.